

PRACTICAL WEEK-2

Session-2

Task1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
GNU nano 8.7.1                                file_copy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source_fd, dest_fd;
    char buffer[1024];
    ssize_t bytes_read, bytes_written;

    source_fd = open("source.txt", O_RDONLY);

    if (source_fd == -1) {
        perror("Error opening source file");
        return 1;
    }

    dest_fd = open("destination.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (dest_fd == -1) {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    while ((bytes_read = read(source_fd, buffer, sizeof(buffer))) > 0) {

        bytes_written = write(dest_fd, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(source_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytes_read == -1) {
        perror("Error reading source file");
    }

    close(source_fd);
    close(dest_fd);

    printf("File copied successfully.\n");

    return 0;
}
```

```

madhulika@LAPTOP-D2M8NKBH:~/OSSP$ mkdir -p practical/week2/task1
madhulika@LAPTOP-D2M8NKBH:~/OSSP$ cd practical/week2/task1
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ pwd
/home/madhulika/OSSP/practical/week2/task1
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ ls
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ nano file_copy.c
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ echo "This is a sample file
  for demonstrating Linux system calls." > source.txt
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ cat source.txt
This is a sample file for demonstrating Linux system calls.
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ gcc file_copy.c -o file_cop
y
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ ls
file_copy file_copy.c source.txt
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ ./file_copy
File copied successfully.
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ cat destination.txt
This is a sample file for demonstrating Linux system calls.
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ ls -l
total 28
-rw-r--r-- 1 madhulika madhulika  60 Aug  9 16:34 destination.txt
-rwxr-xr-x 1 madhulika madhulika 16216 Aug  9 16:34 file_copy
-rw-r--r-- 1 madhulika madhulika 1082 Aug  9 16:33 file_copy.c
-rw-r--r-- 1 madhulika madhulika  60 Aug  9 16:33 source.txt
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task1$ |

```

Observation: The program successfully copied the contents of source.txt to destination.txt using the open(), read(), write(), and close() system calls. The contents of both files were verified to be identical. During execution, the user-space program requested file operations from the Linux kernel through system calls, and the kernel performed the required file operations before returning control to the user-space program.

Task2: Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.


```

+++ exited with 0 +++
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task2$ strace -o trace.txt cat sam
ple.txt
Hello, this is a sample file for system call tracing.
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task2$ ls
sample.txt  trace.txt
madhulika@LAPTOP-D2M8NKBH:~/OSSP/practical/week2/task2$ cat trace.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc740e11d8 /* 26 vars */) = 0
brk(NULL)                               = 0x5c822091e000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7647b7
2b9000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20763, ...}) = 0
mmap(NULL, 20763, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7647b72b3000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) =
 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7647b7280000
mmap(0x7647b7287000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x7000) = 0x7647b7287000
mmap(0x7647b72a8000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28
000) = 0x7647b72a8000
mmap(0x7647b72af000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRIT
E, 3, 0x2f000) = 0x7647b72af000
mmap(0x7647b72b1000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOU
S, -1, 0) = 0x7647b72b1000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3

```

Observation: The strace utility successfully traced the system calls generated by the cat sample.txt command. The trace showed openat() being used to open sample.txt, followed by file and terminal operations using splice(), pipe(), and fcntl(). The close() system call was used to close the file descriptors, and exit_group(0) indicated successful program termination. The observed system calls demonstrate how the cat command interacts with Linux kernel services to access and display file contents.